

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0501

Evergreen Park 230 kV
November 18, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0501. The Interconnection Customer proposes a 260.0 MW wind facility interconnecting at the Evergreen Park 230 kV substation in the Illinois area. The request was received on July 21, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 253 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,441,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Corners / Evergreen Park 230 KV
Contingency: Fisk St / Evergreen Park 230 KV
Loading: 119.2% (Rating: 612 MVA)
DFAX: 44.9%
Pre-project loading: 100.2%
2. Facility: Myrtle Pt / Galion Generating Station 230 KV line
Contingency: Trego Substation / Beaver 500 KV
Loading: 106.2% (Rating: 828 MVA)
DFAX: 7.5%
Pre-project loading: 103.8%
3. Facility: Washington Park / Fisk St 230 KV
Contingency: Fisk St / Evergreen Park 230 KV
Loading: 104.5% (Rating: 545 MVA)
DFAX: 14.4%
Pre-project loading: 97.6%
4. Facility: James De Young / Carley Rd Substation 230 KV
Contingency: E Methuen Substation 500-230 kV TX
Loading: 102.6% (Rating: 463 MVA)
DFAX: 7.2%
Pre-project loading: 98.6%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Evergreen Park / Corners 230 KV Line	Rebuild line with higher-capacity conductor	\$13,084,000	13% (\$1,689,000)
Myrtle Pt / Galion Generating Station 230 KV line	Reconductor line	\$10,260,000	2% (\$222,000)
Washington Park / Fisk St 230 KV	Reconductor line	\$3,034,000	12% (\$367,000)
James De Young / Carley Rd Substation 230 KV	Reconductor line	\$7,481,000	2% (\$163,000)
	Total allocated to project		\$2,441,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.